NOT MEASUREMENT SENSITIVE

MIL-T-13300E(AT)
22 February 1991
SUPERSEDING
MIL-T-13300D(AT)
10 October 1985

MILITARY SPECIFICATION

TRUCKS, HOPPER: COAL DELIVERY, CONVEYOR TYPE, DIESEL ENGINE DRIVEN, 24,000 GVW, 4x2 AND 39,500 GVW, 6x4, MODIFIED COMMERCIAL

This specification is approved for use by the Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers 4x2 and 6x4 diesel engine driven commercial conveyor type coal delivery trucks. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. Vehicles shall be of the following sizes, as specified (see 6.2):

Size I - 24,000 GVW, 4x2
Size II - 39,500 GVW, 6x4

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000 by using the self-addressed Standardization Document Improvement Proposal Form (DD Form 1426) appearing at the end of this document or by letter.

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FSC-2320

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* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	VV-F-800	- Fuel Oil, Diesel.

MILITARY

*	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles, Treatment, Painting, Identification Marking and Data Plate Standards.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

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DEPARTMENT OF TRANSPORTATION (DOT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

* (Application for copies of DOT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations 40 CFR and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, Inc.
SAE Standards and Recommended Practices.

J245	- Engine Rating Code - Spark Ignition.
J516	- Hydraulic Hose Fittings (DOD adopted).
J517	- Hydraulic Hose.
J537	- Storage Batteries (DOD adopted).
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw.
J683	- Tire Chain Clearance - Trucks, Buses and Combinations of Vehicles (DOD adopted).

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J688	- Truck Ability Prediction Procedures (DOD adopted).
J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DOD adopted).
J816	- Engine Test Code - Spark Ignition and Diesel.
J931	- Hydraulic Power Circuit Filtration.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book

(Application for copies of The Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.7, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.7, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

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3.1.1 Special requirements. In addition to the standard vehicle and accessories specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.4 Drain plugs. Drain plugs installed in manual transmissions and in rear axles shall be of the permanent magnet type.

* 3.1.1.5 Towing device. Not less than two hooks, loops, or pins, or a single center mounted towing eye for towing the vehicle shall be furnished on the front of the vehicle. When specified (see 6.2), not less than two hooks, loops, or pins shall be furnished on the rear. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.6 Splash guards. Rear wheels shall be provided with rigid splash guards in front and rubber mudflaps at the rear. A metal strip not less than 1/8-inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or the bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1-1/4 inch wide by 3/32 inch thick shall be furnished at each bolt. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

* 3.1.1.7 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

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3.2 General design.

* 3.2.1 Federal Safety Regulations. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards and Federal Motor Carrier Safety Regulations in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab with all attachments, accessories, and equipment; the body; and a full complement of fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of the curb weight, operator weight (computed at 175 pounds), and a payload to provide not less than the specified GVW.

3.2.6 Weight distribution. Distribution of the GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with a payload uniformly distributed over the load area.

* 3.2.7 Ratings. Component and vehicular ratings shall not be raised to meet the requirements of the specification. The minimum GVW rating shall be not less than 24,000 pounds for size I and 39,500 pounds for size II. Ton ratings specified herein are considered as volume ratings based on 40 cubic feet per ton.

* 3.2.8 Overall width. The overall width of the vehicle, exclusive of tires and safety related items such as mirrors, lights and reflectors shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

* 3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.10 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

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3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. Size I vehicles shall maintain a speed of 50 miles per hour (mph) on grades up to 1.0 percent. Size II vehicles shall maintain a speed of 50 mph on grades up to 0.50 percent. Gradeability requirements shall be met with the main transmission in direct drive and, when a multispeed axle, multirange transmission, or auxiliary transmission is furnished, with the axle, multirange transmission, or auxiliary transmission in high range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. The low speed shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed and shall be not more than 5 mph for size I vehicle and 3 mph for size II vehicle.

3.3.1.3 Maximum speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to the geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Engine governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle size shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle, diesel engine, with not less than six cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

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* 3.4.1.2 Oil filter. A full flow or combination full flow and bypass oil filter with replacement element shall be furnished.

3.4.1.3 Governor. An engine governor shall be furnished, set, and sealed to limit the engine to the engine manufacturer's recommended operating speed.

* 3.4.1.4 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level.

* 3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.7 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.8 Power plant heaters and fuel warmer. When specified (see 6.2), coolant heater, engine oil heater, and fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac) and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

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(a) A coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower, coolant, inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:

(1) Not less than 22 watts per quart heating capacity
(2) Not more than 18 watts per square inch heating capacity
(3) Thermal balance design or thermostat control providing for uninterrupted operation
(4) Provision for mounting below minimum service oil level.

(c) The battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

(d) A fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

* 3.4.1.9 Fuel fired engine preheater. When specified (see 6.2), a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and circulating pump; and shall be connected directly across the engine coolant inlet and outlet. The heater shall be capable of starting and operating at minus 60°F and shall heat the engine to plus 40°F from minus 60°F in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. A 12- or 24-volt direct current (dc) starting system, with 12-volt dc lighting system, shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. Measured shot type ether systems shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for an ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

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* 3.4.2.1.1 Alternator. Unless otherwise specified (see 6.2), a minimum 65-ampere alternator shall be furnished. The alternator output with the engine at engine idle speed shall be not less than 45 amperes.

3.4.2.2 Lighting. All vehicle lights, reflectors, and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19, 393.20, and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on the vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

* 3.4.2.3 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 class A and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have visible and audible type flash indicators.

* 3.4.2.4 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and the total cold cranking ampere rating at 0°F shall be not less than specified in table I. Reserve capacity and cold cranking ratings shall be in accordance with SAE J537. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE I. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel (less than 220 gross horsepower)	320	1200
Diesel (220 gross horsepower and over)	480	1800

* 3.4.2.5 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tanks. Fuel tanks shall have a total capacity of not less than 50-gallons (43 gallons for vehicles destined for California). When more than one tank is furnished, means shall be provided to assure an equalized fuel level in both tanks.

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3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. If vertical exhaust mufflers are furnished and if they are capable of being reached easily by personnel entering or leaving either side of the cab, a heat shield shall be provided. Vertical exhaust systems shall be provided with a hinged rain cap.

* 3.4.5 Transmission. A manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The vehicle shall be provided with not less than 5 forward speeds for size I vehicles and not less than 9 for size II vehicles. When more than five forward speeds are required, a multispeed transmission, auxiliary transmission or two-speed rear axle shall be provided. Gear ratios in the transmission and axles shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide required vehicle performance (see 3.3.1 through 3.3.1.3). The transmission shall be provided with SAE J704 power takeoff openings at right and left sides. The transmission shall provide for maximum ease of shifting in all speeds.

* 3.4.5.1 Clutch. The clutch shall be the largest capacity clutch offered for the size vehicle and engine furnished, with clutch torque capacity exceeding maximum delivered engine torque. The clutch lining shall be asbestos free.

3.4.5.2 Power takeoff. A power takeoff shall be provided. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be manufacturer's standard for the vehicle furnished. The vehicle shall have a frame with reinforcements extending at least from the rear of the front suspension rear hanger bracket to the front of the rear spring front hanger bracket. Reinforcements shall provide sufficient structural strength in the chassis frame to adequately distribute and support the loads imposed when the vehicle body is loaded to specified GVW.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded to specified GVW. When spring capacity is rated at the spring pads, the unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

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* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with vehicle loaded to specified GVW. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

* 3.4.9.1 Two-speed axle. When specified (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. The gear ratio shall provide performance specified in 3.3.1 through 3.3.1.3.

* 3.4.9.2 Rear bogie. The rear bogie on size II vehicles shall be of the four-wheel type complete with axles, springs, torque rods, and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles and with a manually controlled lockout assuring equal power to each rear axle. Lockout control shall be located in the truck cab. Axle gear ratios shall provide performance specified in 3.3.1 through 3.3.1.3.

* 3.4.10 Wheels, rims, tires, and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in procurement documents, multi-piece rims shall not be furnished. Rims and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating) shall be the same for all tires on each vehicle. When specified (see 6.2), wide base type tires and wheels for the front and rear axles shall be provided in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Tires shall be steel belted radial or, when specified (see 6.2), bias ply. Tires shall be tubeless type with highway tread. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the vehicle loaded to specified GVW. Tires and tire size designation systems shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Inner tubes. When tube type tires are specified in acquisition documents, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

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* 3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable safe removal and installation of the spare wheel assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly. The carrier design shall enable removal or mounting of spare wheel assembly using only the tools specified in 3.4.16.1.

* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare wheel or rim or a spare tire assembly is furnished on a vehicle without a carrier, it shall be stowed securely on the vehicle for shipment. When a spare tire assembly is specified, a spare wheel or rim shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design, and load range (ply rating) as furnished on the front axle of the vehicle.

3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

* 3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7.25 cubic feet per minute (cfm)
(b) Air storage reservoir(s), each tank equipped with a drain and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes.

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* 3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished) and additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. When specified (see 6.2), a tilt cab with tilting and locking mechanism shall be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with external, key-operated lock. Full length drip rails shall be installed above the cab doors. Safety grips or grab handles shall be provided on each side of the cab to assist personnel in entering and leaving the cab. When step height into the cab exceeds 24 inches, a secondary step shall be provided. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished. For tilt type cabs, provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield.

* 3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full-width, adjustable seat and back, or individual, adjustable, driver's seat and individual passenger seat. When specified (see 6.2), an individual adjustable driver's seat and an individual passenger seat shall be provided. When specified (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Two pairs of seat belts shall be installed.

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3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by air or electric motor(s).

3.4.15 Bumper. Unless the front bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided. Rear end protection shall be provided and shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate the vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. The stowage space shall provide for positive retainment of this equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle, and wheelnut wrench. The jack shall be of such closed height as to permit its location under an axle or other satisfactory lift point at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to defroster louvers. The heaters shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to driver when seated in driving position.

* 3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

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(a) Key-operated ignition switch
(b) Ammeter, charging indicator, or voltmeter
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light
* (e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
* (h) Ash receptacle
(i) Dual sunvisors
* (j) Front door mounted armrest on driver's side
(k) Driver's compartment ventilator other than window
(l) Tachometer.

When specified (see 6.2), an ammeter or voltmeter gage, an oil pressure gage, and a coolant temperature gage shall be provided in lieu of indicators and buzzers specified above. When specified (see 6.2), an automatic engine shutdown system also shall be furnished. The system shall include an engine coolant temperature, engine coolant level and engine oil pressure red indicating warning light and alarm buzzer. The warning light and alarm buzzer actuation shall precede engine shutdown. The system shall permit engine restart and run for approximately 30 seconds following automatic shutdown. Also, engine oil pressure and engine coolant temperature gages shall be provided with the automatic engine shutdown system.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall have flat and convex areas. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 25 square inches of reflective area and a radius of curvature of not less than 20 inches. The mirrors shall have not less than two supporting arms.

3.4.21 Horn. A manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 Back-up alarm. The vehicle shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

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* 3.4.24 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

3.5 Body. The vehicle shall have the dimensions specified in table II. Cab to rear axle or trunnion dimension may be reduced not more than 2 inches when the vehicle is furnished with a tilt type cab.

TABLE II. Dimensions.

Characteristic	Size I	Size II
Cab to axle (inches-minimum)	84	--
Cab to trunnion (inches-minimum)	--	102
Hopper dimensions		
Inside length (feet-maximum)	12	16
Inside width, overall (feet-maximum)	7	7
Height from chassis frame (inches-maximum)	55	65
Width of running boards (inches, ± 1)	17	17

3.5.1 Body hopper. The body shall be constructed of high-tensile alloy steel sheet of not less than 13 gage (0.0897 inch) thickness. The rear panel shall be sufficiently rigid to insure easy operation of the discharge gate when the truck is fully loaded. The body shall be adequately reinforced so that the rotating jib or lifting device will not cause deformation when lifting the conveyors. The struck (water level) coal capacity shall be computed on the basis of 40 cubic feet per ton.

3.5.1.1 Body compartment. The body shall be of the single compartment type having an open top and a struck water level capacity of not less than 200 cubic feet (5 ton), for size I and 400 cubic feet (10-ton), for size II.

3.5.1.2 Running boards. Two running boards, one on each side of the body for storage of portable conveyors, shall be furnished. Positive clamping devices for holding the conveyors firmly in place during transit shall be provided. The running boards shall be made of safety tread steel plates of not less than 12 gage (0.1046 inch), thickness, exclusive of raised tread. Running boards or support devices shall be adequate to accommodate the outside conveyors specified in 3.5.2.2 when the vehicle is in transit. The outside conveyors shall be capable of being stowed on the vehicle so that they shall not project beyond the rear end of the vehicle, except the 14 foot conveyor in size I may project not more than 2 feet beyond the rear.

3.5.2 Conveyors. One inside and two outside conveyors shall be furnished. The conveyors shall be capable of handling bituminous coal up to an 8-inch maximum size and shall have an unloading rate of not less than 1/2 ton per minute. The body manufacturers recommended engine operating speed (rpm) for operating the conveyors shall be posted inside the vehicle cab.

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3.5.2.1 Inside conveyor. The inside conveyor shall be built into the trough of the body. The conveyors shall be power take-off driven and shall be of the chain-and-flight type. The drag chain of the chain-and-flight type shall have a tensile strength of not less than 16,000 pounds per strand. For the belt-type conveyor, the minimum requirements for the belt shall be four plies of 28-ounce duck with a neoprene coating of 1/16 inch on the top and 1/32 inch on the bottom. Steel flights, connections, and rollers in the flight-type conveyor shall be of comparable weight, strength, and construction. All types of conveyors shall be protected by shields or flashing to prevent coal from working underneath the conveyor surface. Veil plates shall be furnished to prevent the load from bearing directly on the conveyor and shall be individually removable from each compartment when loaded. The veil plates shall be of mild steel, not less than 1/8 inch thick.

3.5.2.2 Outside conveyors. Two outside conveyors shall be furnished. One shall be a nominal 12 feet in length and the other a nominal 14 feet in length. The outside conveyors shall be of the belt type and shall be power driven. The conveyors shall operate from the rear of the truck within a horizontal arc of 180 degrees and at any vertical angle up to a discharge height of 7 feet from the ground. The conveyors shall lock securely in position and engage the driving mechanism positively under all operating conditions. The belts shall be not less than 10 inches wide and the minimum requirements for the material shall be three plies of 28-ounce duck with a neoprene coating of 1/16 inch on the top and 1/32 inch on the bottom. Nonslip surfacing, if used, shall be of nonabsorbent material which will not freeze in cold weather.

3.5.2.3 Power takeoff. A power takeoff equipped with self-alining bearings shall be installed for driving the inside and outside conveyors from the truck engine. Controls for the operation of the unloading shall be located at the rear curbside of the truck.

3.5.2.4 Driving mechanism. The driving mechanism shall be driven by the power takeoff. The mechanism shall be either a mechanical type such as gear or chain or a hydraulic type. All intermediate and universal joints, transmissions, worms, gears, pinions, shafting, bearings, chain, sprockets, pulleys, and other parts required to make a complete installation shall be furnished. All moving parts which may be a hazard to personnel shall be adequately guarded.

3.5.2.5 Hydraulic driven mechanism. When a hydraulic driven system is furnished, it shall consist of a pump unit, hydraulic motors driving inside and outside conveyors, control valve, fluid reservoir, hoses, couplings and all other parts essential for normal operation. The system shall incorporate a pressure relief valve to prevent buildup of pressure exceeding the rating of any component. Hydraulic hose shall be single wire braid or double wire braid, rubber covered, conforming to SAE J517. Hose fittings shall conform

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to SAE J516. The hydraulic system shall incorporate a filtration system conforming to SAE J931. A corrosion-resistant plate indicating the type of hydraulic fluid to be used shall be installed near the filler cap.

3.5.2.6 Conveyor handling device. A conveyor handling rotating jib or lifting device shall be furnished. The handling device shall be capable of placing and holding the outside conveyors in all positions throughout their operating range (see 3.5.2.2).

3.5.3 Mounting. The body shall be mounted with U-bolts, twin studs, or chassis frame rail brackets. The vehicle chassis frame shall be braced using a wood block at each U-bolt or stud mounting point unless the mounting point is located at a full depth, frame crossmember. Blocks shall incorporate a keeper strap or groove for the mounting bolt, and shall be of a width to assure retention. On U-bolt or stud mounted bodies, two shear bolts shall be provided, one on each side of the rear of the body subframe to maintain body alinement on the chassis. When the body is secured by brackets attached to the chassis frame rails, attachments shall be made to only the web of the frame rails. When additional holes are required to secure mounting to chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

* 3.5.4 Identification and instruction plates. All body operating controls shall be provided with identification plates and, as appropriate, with caution, warning and instruction plates. Plates shall conform to Composition A (class 1 or 2) or Composition C of MIL-P-514.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: focusing of lights, adjustment of engine, adjustment of electrical and brake systems, filling and charging of batteries, alinement of front wheels, inflation of all tires, complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point, servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume, and servicing of the windshield washer reservoir with water and appropriate additives.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Vehicle weights. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on the front and rear axles. The total imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds and the payload required to provide the specified GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle, and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

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4.3.2 Road and functional tests. The first production vehicle shall be examined and road tested (less payload) by the contractor to assure that the vehicle will operate in accordance with contract requirements. All functional operations of the body shall be demonstrated under loaded conditions.

* 4.3.3 Body treatment and painting. The certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to the Government representatives at the first production vehicle inspection.

4.3.4 Production sample. Upon acceptance of the first production vehicle, the vehicle shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.5 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall as a minimum assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government in transporting coal.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of vehicle required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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(d) Identification of appropriate service for painting and marking (see 3.1.1.1).
(e) Concealed marking, if required (see 3.1.1.1).
* (f) Rustproofing, if required (see 3.1.1.3).
(g) Towing devices on rear, if required (see 3.1.1.5).
* (h) Satisfactory operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).
(i) Silicone rubber hoses, if required (see 3.4.1.6).
(j) Power plant heaters, if required (see 3.4.1.7).
(k) If a battery heater is not required (see 3.4.1.7).
(l) Fuel fired engine preheater, if required (see 3.4.1.8).
* (m) Alternator capacity if not 65 amperes (see 3.4.21).
(n) Two-speed axle, if required (see 3.4.9.1).
(o) Wide base type tires and wheels, if required (see 3.4.10).
(p) Disc type wheels, if required (see 3.4.10).
(q) Bias ply or other type tires, if required (see 3.4.10.1).
* (r) Low profile tires, if required (see 3.4.10.1).
(s) Spare tire carrier, if required (see 3.4.10.3).
(t) Spare wheel or rim, if required (see 3.4.10.4).
(u) Spare tire assembly, if required (see 3.4.10.5).
(v) Air dryer, if required (see 3.4.11.1).
(w) Brake controls for use from a towing vehicle, if required (see 3.4.11.2).
(x) Tilt type cab, if required (see 3.4.12).
* (y) An individual adjustable driver's seat, if required (see 3.4.12.1).
* (z) An air ride driver's seat, if required (see 3.4.12.1).
(aa) Tools, if required (see 3.4.16.1).
* (ab) Gages, if required, in lieu of indicators (see 3.4.19).
* (ac) Automatic engine shutdown, if required (see 3.4.19).
(ad) Air operated horn, in addition, if required (see 3.4.21).
* (ae) An AM/FM radio, if required (see 3.4.24).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type and class. Unless other conditions are cited in the contract computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173

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Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

* 6.4 Subject term (key word) listing.
Body hopper
Chassis truck
Conveyor
Nontactical truck
Truck, commercial
4x2 or 6x4.

* 6.5 Identification of changes. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-T-13300E(AT) TRUCKS, HOPPER: COAL DELIVERY CONVEYOR TYPE, DED, 24,000 GVW, 4x2 AND 39,500 GVW, 6x4, MODIFIED COMMERCIAL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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